CNC-500 Machine Maintenance Manual

1. Hydraulic System

The CNC-500 uses a hydraulic system for machine operation. The hydraulic pump should be inspected every 500 operating hours. The inspection should include checking the oil pressure, oil level, and possible leakage.

If hydraulic pressure drops below the recommended operating range, the machine should be stopped and the hydraulic system should be inspected before continuing operation.

2. Motor Temperature

The motor normally operates between 60°C and 80°C. Temperatures within this range are considered normal during regular operation.

If the motor temperature exceeds 90°C, stop the machine immediately and inspect the cooling system. Check whether the cooling fan is operating correctly and make sure that the ventilation openings are not blocked.

3. Safety Equipment

Employees performing maintenance must wear protective gloves and safety glasses. Safety shoes should also be worn when working around heavy machine components.

Before performing maintenance, the machine must be switched off and disconnected from its power supply.

4. Lubrication

Moving mechanical components should be lubricated every 250 operating hours. Only manufacturer-approved lubricant should be used.

Insufficient lubrication may cause increased friction, component wear, and higher operating temperatures.

5. Vibration Monitoring

During normal operation, the machine vibration level should remain below 5 mm/s. If vibration exceeds 8 mm/s, the machine should be inspected for mechanical problems.

Possible causes of excessive vibration include bearing wear, incorrect alignment, and loose machine components.

6. Emergency Procedure

If abnormal noise, excessive vibration, smoke, or unusually high temperatures are detected, stop the machine immediately.

Disconnect the machine from the power supply and inform the maintenance team before restarting the equipment.

7. Cooling System Inspection

The cooling system should be inspected every 300 operating hours.

The cooling fan should be cleaned if dust blocks the ventilation openings.

This is routine maintenance and does not require stopping the machine

unless the motor temperature is above the safe operating range.

8. High Temperature Procedure If the motor temperature exceeds 90°C, stop the machine immediately. Before restarting the machine, inspect the cooling fan, ventilation openings, and coolant circulation.

9. CNC-600 Motor

The CNC-600 normally operates between 55°C and 75°C.

If its temperature exceeds 85°C, reduce the machine load

and inspect the ventilation system.

10. Employee Heat Safety Employees working in high-temperature environments should take regular breaks and drink sufficient water.